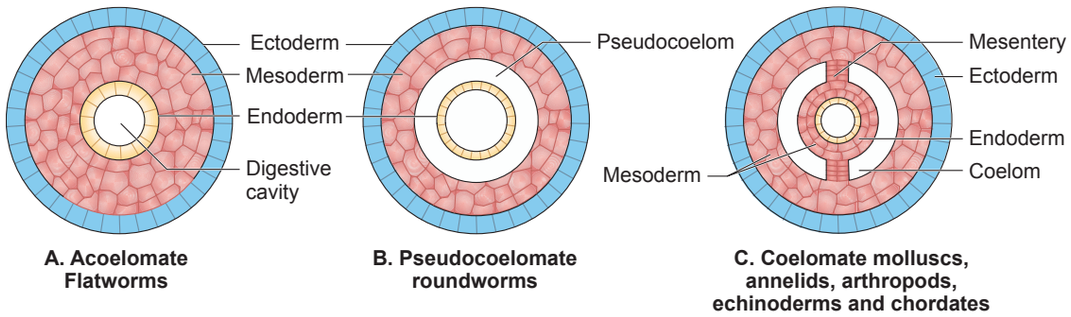


Figure 2.6 Diagrammatic representation of coelom in animals

body cavity is formed by splitting of mesoderm. (e.g., annelids, arthropods, molluscs). In **Enterocoelomate animals** the body cavity is formed from the mesodermal pouches of archenteron. (e.g., Echinoderms, hemichordates and chordates) (Figure 2.7).

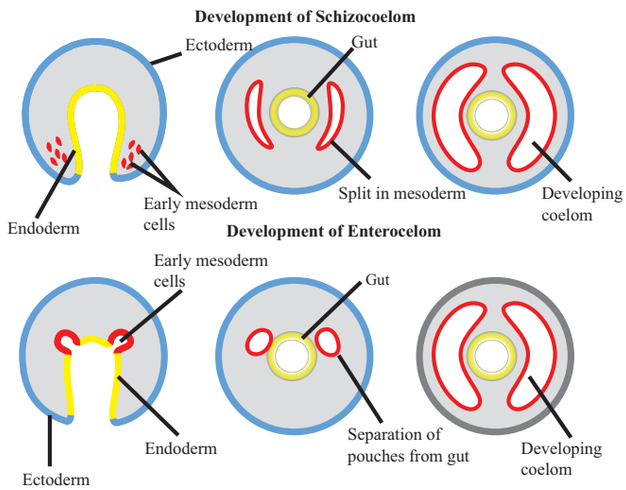


Figure 2.7 Development of Schizocoelomata and Enterocoelomata

What is the advantage of true coelom over a pseudocoelom?

2.1.5 Segmentation and Notochord

In some animals, the body is externally and internally divided into a series of repeated units called segments with a serial repetition of some organs **(Metamerism)**. The simplest form of segmentation is found in Annelids in which each unit of the body is very similar to the next one. But in arthropods (cockroach), the segments may look different and has different functions.

Animals which possess notochord at any stage of their development are called chordates. Notochord is a mesodermally derived rod like structure formed on the dorsal side during embryonic development in some animals. Based on the presence or absence of notochord, animals are classified as chordates (Cephalochordates, Urochordates, Pisces to Mammalia) and nonchordates (Porifera to Hemichordata).

2.2 Classification of Kingdom Animalia

Animal kingdom is divided into two sub-kingdoms, the Parazoa and Eumetazoa based on their organisation.

1. Parazoa: These include the multicellular sponges and their cells are loosely aggregated and do not form tissues or organs.

2. Eumetazoa: These include multicellular animals with well defined tissues, which are organised as organs and organ systems. Eumetazoans includes two taxonomic levels called grades. They include **Radiata** and **Bilateria**.

TN_GOVT_XI_Bio Zoology_Ch02.indd 18 09-01-2020 5.24.01 PM